

Creation Phase

This project began with finding a dataset that could tell a meaningful story for a real business audience. I searched Kaggle for consumer behavior data because I wanted something that could apply directly to a financial marketing standpoint — something a real company could act on. I selected the `shopping_behavior_updated.csv` dataset, which contains 3,900 rows and 18 columns of anonymous consumer purchasing records from an online retail platform. The dataset covers customer demographics such as age and gender, purchase details including category and amount, and transactional attributes like discount usage, promo codes, season, and purchase frequency. It contained no personally identifiable information, meeting the ethical requirements of the assignment. I used Claude as a collaborative design partner throughout the creation process. Rather than simply generating code and submitting it, I used AI to help me think through the architecture of the dashboard, challenge my assumptions about the data, and refine my visual and narrative choices. The collaboration looked like a dialogue — I would describe what I was trying to prove, Claude would suggest approaches, I would test them, and then we would iterate based on what the data showed.

My initial question was broad: 'Who is actually buying and what makes them spend more?' Through conversations with AI, I explored what the data could realistically answer. We identified three core stakeholder questions that would drive the dashboard:

- Which customer segment generates the most revenue?
- Are discounts actually driving more spending?
- Who is being overlooked by current marketing strategies?

As I analyzed the data more carefully, the main question evolved into something more specific and powerful: 'Who should we market to — and is the discount actually worth it?' This shift happened because the data revealed a finding that directly challenged assumptions: 100% of discounts in the dataset went exclusively to male customers, yet female customers spent more per purchase than males without receiving any incentive to do so. This became the central story of the entire dashboard.

I initially explored the data in Power BI, creating basic bar charts and pie charts for age, gender, clothing type, and season — just to establish a baseline understanding of the dataset. However, I made the deliberate decision to switch to Python and Streamlit for the final dashboard for two reasons. First, Streamlit allows free deployment to a live public URL, making the dashboard accessible to anyone without requiring installation. Second, Python and Streamlit carry stronger portfolio appeal on GitHub, which is important for future job applications in data analytics. I set up a Python virtual environment inside a dedicated project folder on my Mac, installed Streamlit, Pandas, and Plotly, and built the dashboard incrementally — testing each chart as it was added. The AI collaboration was particularly valuable during the technical setup phase, helping me navigate Mac-specific installation issues and structure the code in a professional, reproducible way.

The dashboard was designed around three principles: organization for efficiency, directness to reduce confusion, and aesthetic engagement to keep the viewer interested. Every design decision was made with a non-technical marketing manager in mind — someone who needs to make a budget decision, not someone who wants to see statistical methodology. The color scheme uses navy as the background, light blue for neutral data, green for positive findings, and red for warnings and problems. This creates an immediate visual language — viewers instinctively understand that green is good and red is a problem before reading a single word. Chart types were selected based on what each comparison required rather than defaulting to bar charts for everything:

- Bar charts — comparing total revenue across categories and age groups
- Box plots — showing the full distribution of spend to reveal differences that averages hide
- Scatter plots — exploring the relationship between age and purchase amount
- Tree maps — showing part-to-whole relationships for category and gender revenue
- Heatmap — displaying purchase frequency across age groups simultaneously
- Custom progress bars — immediately visualizing the revenue dominance of the 56+ segment

Several significant changes were made during the creation process based on what the data revealed:

- Initial charts used average spend which produced nearly identical bar heights (\$58-\$61 range). Switching to total revenue showed real variation between groups and made the 56+ dominance visible.
- The main question changed when I discovered 100% of discounts went to males — this shifted the entire narrative from a general spending analysis to a targeted critique of discount allocation strategy.
- The tab structure was reorganized around the real story: Overview, Finance, Discounts, Selling Frequency, and Recommendations — each answering one specific question.
- The y-axis on gender and discount box plots was zoomed to start at 30 rather than 0, making distributional differences visible that were previously invisible.
- A promo code chart crashed during testing due to a wrong column name reference and a duplicate variable name — both were identified and corrected through careful debugging.
- The Plotly toolbar was hidden across all charts to create a cleaner, less cluttered presentation experience.

Stakeholder Simulation

Red = AI [claude] ; Black = Ally

Persona from AI: A skeptical Marketing Director at a mid-size online clothing retailer. She controls the quarterly ad and promotions budget and reports directly to the CEO. She is data-literate but not a data scientist — she needs clear, actionable insights and will push back hard on anything that feels vague or unsupported.

Question 1: What is this dashboard actually telling me?

Stakeholder: I opened this dashboard and the first thing I see is a warning box telling me my budget is being spent on the wrong people. That is a bold claim. Walk me through what you mean.

Response: The data shows two things happening simultaneously. Your 56+ male customers make up the most purchases and generate the most total revenue. However, when you look at average spend per transaction, female customers consistently outspend males at \$60.25 versus \$59.54. Now look at the Discounts tab: every single discount and promo code in this dataset went exclusively to male customers. Female customers received nothing and yet they still bought. That means you are spending discount budget on a group whose behavior does not change because of it, while completely ignoring a group that is already your highest value buyer per transaction.

Question 2: The spend difference between genders is less than a dollar. Why should I care?

Stakeholder: You are showing me that females spend \$60.25 and males spend \$59.54. That is 71 cents. I am not restructuring my entire marketing strategy for 71 cents.

Response: The 71cent difference per transaction is not the main point. The main point is that female customers spend more without any incentive; no discounts, no promo codes, nothing. Males who receive discounts spend less than males who do not receive them. So the question is not about 71 cents per transaction. The question is why you are spending discount budget at all when it demonstrably does not increase spending for the group receiving it, and why you are not running any campaigns toward the group that already spends more without needing to be incentivized. The issue is not the size of the difference; it is the direction of your investment.

Question 3: Why is my 56+ customer the target if they don't spend the most per purchase?

Stakeholder: You said females spend the most per purchase. But your recommendations still say focus on 56+ males. That seems contradictory.

Response: These are two different strategies for two different goals. 56+ males are your largest customer base, and they generate the most total revenue simply because there are more of them buying. They are already loyal customers and do not need discounts to return. Female customers are your valuable because they spend more per visit and receive nothing from you currently. The recommendation is not to choose one over the other. It is to stop spending discount budget on 56+ males who would have bought anyway and redirect that into campaigns targeting female customers who are currently invisible to your marketing.

Question 4: What is the single most important thing this tells me to do? What should I get out of this?

Stakeholder: I have five minutes before a budget meeting. Give me one sentence.

Response: Stop giving discounts to customers who would have bought anyway, and start running campaigns toward female customers who already spend more than anyone else without needing a reason to.

AI Stakeholder Critique of the Dashboard:

What Works Well :

- The tab structure is logical and follows a clear narrative from problem identification through to recommendation — a busy executive can navigate it without instruction.
- The warning box on the Overview tab immediately establishes why this research matters — the premise is understood before a single chart is read.
- The progress bars are a unique and effective visual device — the dominance of 56+ is communicated instantly without requiring chart interpretation.
- The Recommendations tab uses plain English with no technical jargon — a marketing manager can take those four boxes directly into a budget conversation.
- The color language is consistent and intuitive — green means positive, red means problem, blue means neutral data.
- The sidebar filters make the dashboard reusable — a marketing manager can isolate specific seasons, categories, or genders to answer their own questions.

What Could Be Stronger:

- The dataset limitation — flat spend distribution suggesting synthetic data — should be acknowledged directly on the dashboard rather than only in this report. A small disclaimer on the Overview tab would build trust with a skeptical stakeholder.
- The Finance tab title 'Where is the money coming from?' could be more specific — something like 'Revenue by Age and Gender' would tell the viewer what they are about to see before they see it.
- The scatter plot showing age versus purchase amount effectively confirms there is no correlation — but a viewer unfamiliar with scatter plots may not immediately understand what a cloud of dots with no pattern means. A brief annotation would help.
- The dataset covers a single anonymous retailer — recommendations could note that findings would need validation against the company's own transaction data before being acted upon.

What I fixed inconclusion to the simulation:

I altered the dashboard accordingly by making mainly two adjustments that were made to the dashboard. First, the gender spending comparison chart had its y-axis zoomed to begin at 30 rather than 0, making the difference between male and female spend visible rather than appearing flat. Second, the discount box plot received the same treatment of zooming to 30 made the distributional difference between discounted and non-discounted customers clear. Both changes were direct responses to the stakeholder's challenge that the differences looked too small to matter and by adjusting the scale, the visual argument became much stronger without changing any of the underlying data.

Reflective Analysis

Before:

The topic of this project is consumer shopping behavior from an anonymous online retail platform. I wanted to understand who is buying, what they are buying, how much they are spending, and whether discounts and promo codes are actually making a difference in how customers shop. The main message is that retail marketers are wasting their discount budget on male customers whose spending behavior does not change because of it, while completely ignoring female customers who already spend more per purchase without any incentive. There is a clear misalignment between where the money is going and where the value actually is.

The audience for this dashboard is marketing managers and e-commerce directors at mid-size online clothing retailers. I narrowed it down to one specific person: a marketing manager who controls the ad and promotions budget and reports to a CEO. This person understands business but is not a data scientist, so they need visuals and language that get straight to the point. They probably know their customer base skews male and that they run seasonal promotions. However, they likely do not know that spending is nearly identical across all age groups, that discounts are not actually driving more revenue, or that female customers outspend male customers per transaction. Most retailers assume younger customers are the best target, and this data challenges that assumption.

A marketing manager might assume that discounts are necessary to drive conversions and that taking them away would hurt sales. They might also assume that because males are the majority customer, they should be the primary target. They may believe younger shoppers spend more because they are more brand driven, or that campaigns should be concentrated in peak seasons rather than run year round. The dashboard is built to support two specific decisions. First, whether to reallocate discount budget away from broad male targeted promotions and toward targeted female focused campaigns. Second, whether to continue blanket discounting at all or redirect that spend into advertising. Every chart was selected and organized to support one of those two decisions.

Female customers spend more per purchase than male customers and receive zero discounts yet they still buy. This is the most actionable finding in the entire dataset because it identifies an untapped marketing opportunity and a budget inefficiency at the same time. The tone needed to be direct and professional without being overly academic. I wanted the dashboard to feel like it was speaking to the marketing manager as a peer rather than lecturing them. Every label and recommendation was written in plain English with no statistical jargon so someone could open the dashboard and immediately understand what the data shows and what they should do about it.

After:

The Overview tab opens with a red warning box that states the core problem before the viewer looks at a single chart. The progress bars show the revenue dominance of 56+ males right away. Then the Finance and Discounts tabs build the evidence step by step. All these make sure the main question at hand is answered directly. The Recommendations tab closes with four plain English action items. The flow was intentional and I feel like it reads clearly from start to finish. The flatness of the spend data across all segments is the biggest weakness. Every group spends between \$58 and \$61 which looks suspicious and could make a skeptical viewer question whether the dataset is real. I did not address this directly on the dashboard itself and looking back

I think adding a small disclaimer on the Overview tab would have helped build trust with someone who is analytically minded.

Yes, the side-by-side comparisons like discount vs no discount and male vs female are the clearest parts of the dashboard and allow the message to be more direct. The box plots improved the analysis a lot because they show the full spread of spending rather than just averages. I also zoomed the y axis on those charts to start at 30 instead of 0 which made the differences actually visible instead of looking completely flat. The color, typography, and annotation was one of my favorite parts of building it and I noticed it overall really did help in reinforcing the main idea. I used green for positive findings, red for warnings or problems, and light blue for neutral data. This color system means a viewer does not need to read every label to understand the direction of a finding because the color tells them before they even get to the text. I also removed the Plotly toolbar from all charts which made everything look a lot cleaner. The dark navy theme looks professional and resembles tools like Tableau or Power BI that marketing professionals already use which is what I was aiming for. The tab structure makes the dashboard easy to navigate without instructions. The recommendations are formatted as cards so they are easy to scan quickly which is important for someone who may be looking at this right before a meeting.

What questions might a skeptical audience member ask?

- The gender spend difference is only 71 cents. Is that actually meaningful in practice?
- Why does the scatter plot show no relationship between age and spending?
- If discounts only go to males, could that just be how the retailer's loyalty program works rather than a strategic mistake?
- The recommendations say to target female customers but what should those campaigns actually look like?

The finding that surprised me most was that spending was nearly identical across every demographic segment. Age, gender, subscription status and discount usage all fell within a \$58 to \$61 range. I expected the data to show clear winners and losers in terms of spending behavior. Instead, the flatness of the data forced me to rethink my central question. Instead of asking what makes customers spend more I had to ask who is being given resources they do not need and who is being ignored despite already performing well. That shift produced a much stronger and more actionable story.

The most important skill I developed through this project was learning how to tell a story when the data does not make it easy. The data did not give me dramatic differences to visualize. It gave me a subtle but structurally important pattern which was the complete exclusion of female customers from discount programs despite their stronger per transaction value. Getting that finding to feel urgent and actionable for a non technical audience required a lot of intentional design decisions around color coding, axis scaling, language and narrative structure across the tabs. If I rebuilt this dashboard, I would look for a dataset with more spend variance to make the visual argument stronger. I used Claude to help me challenge my own assumptions, design decisions and review my coding work when it was crashing.